Machine Learning Models Reveal Key Performance Metrics of Football

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Core Problem

What Determines Winning in Football?

Football matches are influenced by multiple factors, but
it is unclear which player performance metrics truly lead to winning.

Traditional analysis:

- Relies on intuition and basic stats
- Fails to capture complex interactions

Why do we even need it?

Football decisions today are often:

- Based on intuition and experience
- Focused on limited statistics (goals, assists)

But actually:

- Matches depend on multiple hidden factors and statistics
- Player contributions are complex and interconnected

Why do we even need it?

By knowing all those complex patterns about the game we can

- Identify what truly impacts winning
- Go beyond basic stats
- Make data-driven decisions

majorly all the football clubs earn via ticket sales, sponsorship and some other deals
All of those will help in revenue generation for running a club, and they all depend on a crucial factor, i.e., how the team actually performs and how well they do the business of buying and selling the players

- Clubs like Dortmund, Ajax, Lyon do not have wealthy owners. They operate on self-sustaining, revenue-based models
- Cannot compete in high-budget player transfers so they rely heavily on data-driven scouting and analytics.
- Use key performance metrics to identify undervalued talent
- Focus on smart, cost-effective player recruitment.

Such models are also used by major companies to evaluate and select club sponsorship deals based on team performance. Companies leverage these models to identify high-potential clubs for sponsorship by analyzing performance data.

Dataset Overview and Feature Engineering

800+ matches, 2000+ players, Data from Qatar Stars League (2012–2019)

The dataset is divided into 4 categories:

- Player Information: Player ID, team, opponent, position.
- Match Information: Match result, season, location.
- Technical Performance Shots, passes, tackles, fouls, etc.
- Physical Performance: Speed, distance covered, movement intensity

Dataset Overview and Feature Engineering

Feature Engineering Approach:

- Players grouped into:
  - Forward | Midfield | Defense
- Technical metrics:
  - 16 metrics aggregated (sum) per position
  - 48 features (eg:- successful_passes_fwd, mid , def).
- Physical metrics:
  - 6 metrics aggregated (mean) per position
  - 18 features
- Player distribution:
  - Number of players per position
  - 3 features (no. of player in def, mid and attack)

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Why we need to do feature engineering

- Capture position-wise performance
- Combine individual stats into team-level insights

Models taken into account

Machine Learning Models Applied:

- Logistic Regression (LR)
- Random Forest (RF)
- Decision Tree (DT)
- Support Vector Machine (SVM)
- k-Nearest Neighbors (kNN)
- Multilayer Perceptron (MLP)
- XGBoost

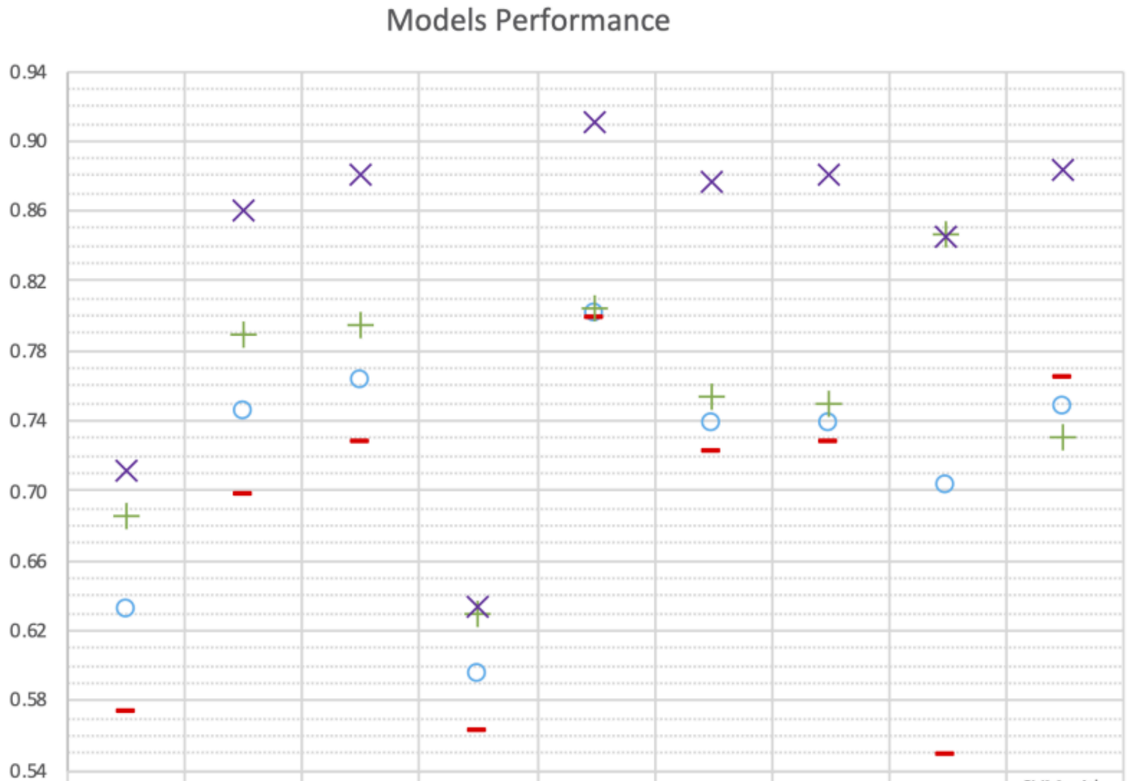
Feature seleciton techniques

It is a must done, to remove the irrelevant features which were present in our data
Techniques Used:

- Pearson Correlation Coefficient (PCC)
- Decision Tree (Gini Index)
- Random Forest Importance
- Logistic Regression Coefficients
- Recursive Feature Elimination (RFE)

Number of Features Selected: 31
Total features: 69
Best performance with: 31 features (RFE)
Final model used: All 69 features (slightly better accuracy)

Performace



	K Nearest Neighbors	Random Forest	XG Boost Classifier	Decision Tree	Logistic Regression	MLP Classifier	SVM with linear kernel	SVM with RBF kernel	SVM with polynomial kernel
Accuracy	0.6320	0.7450	0.7630	0.5960	0.8012	0.7390	0.7390	0.7030	0.7480
Sensitivity	0.5740	0.6980	0.7280	0.5620	0.7989	0.7220	0.7280	0.5490	0.7650
Specificity	0.6860	0.7890	0.7940	0.6290	0.8037	0.7540	0.7490	0.8460	0.7310
AUC	0.7120	0.8600	0.8800	0.6340	0.9114	0.8770	0.8810	0.8450	0.8830

LR performs best, followed by XGBoost and Random Forest

What actually wins matches?

1. Shots on Target (Forwards)
   - Strongest indicator of winning
   - More shots, higher scoring probability/More chances created.

2. Successful Passes (All Positions)
   - Forward, Midfield, Defence
   - Better passing → better posession

3. High-Speed Distance
   - Players covering more distance at high speed
   - Indicates fitness + intensity.
   - It’s a team game at times, and at times the winner is just decided by pure physical attributes.

4. Role of Defenders
- Clearances
- Successful passes
- Even shots on target

5. Fair Play Matters
- When teams have fewer yellow/red cards they have a higher chance of winning
- Players on a yellow card cannot afford another mistake to get sent off
- overall, team comes under pressure

Winning depends on attacking efficiency, passing accuracy, physical fitness, and disciplined gameplay

What should a team do?

How can teams use these insights?

Specialised training plans to improve specific attributes of the player such as:-

- Improve shot and passing across all positions
- Increase player fitness and strengthen defensive contribution
- Maintain discipline (fair play)

and many more such insights

Besides that, it could be used for scouting the right players according to the team’s performance and issues

What was missing?

It doesn’t take injuries into account, which is the major factor for predicting a tie, at times. At times, key players are missing during important matches, leading to a loss. Also Injury record help in knowing where the team is lacking in fitness and what actions could be taken upon that

The model does not consider player identity or the presence of key players, focusing only on aggregated team metrics. This means it fails to capture individual brilliance, where a single player can influence the outcome of a match even if the overall team performance is poor.

Draws and goalkeeping metrics were not included in the analysis, even though they may have a significant influence on match outcomes.

Thankyou